Listing 4.6 Use the NASA GIS RESTful API to retrieve an image of lake Kariba taken by VIIRS from the previous day.

```
import requests
from datetime import datetime, timedelta

bbox = (26.5, -18.5, 29.0, -16.0) #1
yesterday = (datetime.now() - timedelta(1)).strftime('%Y-%m-%d')

params = {
    'SERVICE': 'WMS',
    'VERSION': '1.1.1',
    'REQUEST': 'GetMap',
    'LAYERS': 'VIIRS_SNPP_CorrectedReflectance_TrueColor',
    'SRS': 'EPSG:4326',
    'BBOX': f'{bbox[0]},{bbox[1]},{bbox[2]},{bbox[3]}',
    'WIDTH': '1024',
    'HEIGHT': '640',
    'FORMAT': 'image/png',
    'TIME': yesterday,
    'TRANSPARENT': 'true'
}

response = requests.get( #2
  'https://gibs.earthdata.nasa.gov/wms/epsg4326/best/wms.cgi',
  params=params)

filename = f'lake_kariba_{yesterday}.png'
with open(filename, 'wb') as f:
    f.write(response.content)
```

#1 Define the area of interest on the surface of the earth as a bounding box using EPSG:4326 (WGS*4) coordinates.
#2 Query the RESTful API with an HTTP GET request.

4.4.4 ETL and data transformation

Extract, *transform*, *and load* (ETL) processes bridge the gap between diverse source systems and standardized digital twin storage. ETL operations ensure data consistency regardless of source format, apply business rules and validation, and aggregate and optimize data for analytical queries.

The transformation stage handles data cleansing, type conversion, aggregation, and enrichment. Raw sensor readings might be validated against expected ranges, converted to standard units, and enriched with contextual information from reference data systems.

TRY IT OUT: USE APACHE SPARK TO AGGREGATE AND TRANSFORM POWER CONSUMPTION DATA

The optical sensor I fitted to my main electrical meter in chapter 3 allows me to export minute by minute power consumption data as CSV. I would like to aggregate that data into